Assignment 3

Due date: TBA.
Total: 30 points
Coverage: Chapter 4 (Generating Random Variables).

Question	Format	Points
1	8 multiple-choice questions on inversion and rejection sampling	8
2	8 multiple-choice questions on constructing and checking random variables	8
3	6 true/false questions	6
4	2 short explanations with four prompts each	8
Total		**30**

Assignment PDF | Response template (.qmd)

Submission

Use **R Markdown** or **Quarto**. Submit both your **source file** (`.Rmd` or `.qmd`) and its **rendered PDF** to the **Assignment 3** folder on **iCollege**. Use the filename `A3_LastName_FirstName` with the appropriate extension and include your name in the PDF.
Begin the PDF with an answer table for Questions 1-3, followed by Question 4. Use the supplied response template, or make your own table with the same part labels.

- **Questions 1-2:** Select one best answer, A, B, C, or D.
- **Question 3:** Write **True** or **False**.
- **Question 4:** Explain your reasoning in your own words, using one or two sentences per numbered prompt.

Only Question 4 requires explanations. **No code execution is required.** R code and figures are optional. If you use R to check your work, include the code in your source file. The submitted source must render to the submitted PDF without errors.

The questions use the main material in Chapter 4. Topics marked **optional**, including Box-Muller, truncated-normal simulation, Wishart construction, and the real-world applications, are not required.

Notation: $N(\mu, \sigma^2)$ uses variance as its second parameter. Gamma(a, b) uses shape a and **rate** b. Use the independence assumptions stated in each question.

Question 1: Uniform Inputs, Inversion, and Rejection [8]

Select the **best answer**. Each question is worth **1 point**.

1a)

Consider the following R code:

```
x <- numeric(100)
for (i in seq_len(100)) {
  set.seed(8670)
  x[i] <- runif(1)
}
```

What is the main problem with using `x` as 100 independent uniform draws?
A. The values must all be zero.
B. The values are sampled without replacement.
C. Resetting the seed repeats the same first draw in every iteration.
D. A uniform generator cannot be used inside a loop.

1b)

Let $U \sim \text{Unif}(0, 1)$. Which expression generates an exponential random variable with **mean 5**?
A. $-\log(1-U)/5$
B. $-5\log(1-U)$
C. $\exp(-5U)$
D. $5U$

1c)

A target density is $f(x) = 2(1-x)$ for $0 < x < 1$, and zero otherwise. Which inverse-transform generator has this density? Use $U \sim \text{Unif}(0, 1)$.
A. $X = \sqrt{U}$
B. $X = 1 - U$
C. $X = U^2$
D. $X = 1 - \sqrt{1-U}$

1d)

A discrete random variable takes values -2, 1, and 5 with probabilities 0.20, 0.50, and 0.30, respectively. Using $F^{-1}(u) = \inf\{x : F(x) \geq u\}$, what value is returned at the exact probability level $u = 0.70$?
A. 1
B. -2
C. 5
D. No value is defined at a jump.

1e)

A continuous, strictly increasing CDF on $[0, \infty)$ is inverted numerically. For a draw $u = 0.90$, a proposed search interval $[a, b]$ has $F(a) = 0.25$ and $F(b) = 0.75$. What should be done before applying a bracketing root finder to $F(x) - u$?
A. Accept the interval because both CDF values lie between 0 and 1.
B. Replace the random draw by 0.75.
C. Increase the upper endpoint until $F(b) \geq 0.90$.
D. Change the stopping tolerance without changing the interval.

1f)

A rejection sampler targets $f(y) = 2y$ on $(0, 1)$, uses a uniform proposal $g(y) = 1$, and takes $M = 2$. For one independent proposal/decision pair, $Y = 0.80$ and $U = 0.65$. What should the sampler do?
A. Accept and store 0.80.
B. Accept and store 0.65.
C. Reject because $0.65 < 0.80$.
D. Accept and store 1.60.

1g)

A rejection sampler uses normalized target and proposal densities with a valid bound $M = 2.5$. What is the **expected** number of proposals needed to obtain 400 accepted observations?
A. 160
B. 400
C. 800
D. 1000

1h)

A programmer chooses a rejection bound by evaluating $f(y)/g(y)$ at 100 grid points and taking the largest observed ratio. Which step is needed to justify the sampler's correctness? Assume the proposal covers the target support.
A. Increase the number of accepted observations.
B. Establish that the bound holds throughout the target support.
C. Replace every acceptance ratio above 1 by 1.
D. Choose a different random seed.

Question 2: Constructing and Checking Random Variables [8]

Select the **best answer**. Each question is worth **1 point**.

2a)

Let $X_1 \sim \text{Poisson}(1.2)$ and $X_2 \sim \text{Poisson}(2.8)$ be independent. What is the distribution of $X_1 + X_2$?
A. Poisson(1.6)
B. Poisson(3.36)
C. Poisson(4)
D. Binomial$(4, 0.5)$

2b)

Suppose $G_1 \sim \text{Gamma}(2, 4)$ and $G_2 \sim \text{Gamma}(5, 4)$ are independent, with the second parameter denoting the **rate**. What is the distribution of $G_1/(G_1 + G_2)$?
A. Beta$(2, 5)$
B. Beta$(5, 2)$
C. Gamma$(7, 4)$
D. Unif$(0, 1)$

2c)

Let $Z_1, \ldots, Z_6$ be independent standard normals. A programmer returns

$$W = \frac{Z_1}{\sqrt{(Z_1^2 + \cdots + Z_6^2)/6}}$$

and calls the result a t_6 draw. Which statement identifies the error?
A. The sum of six squared standard normals cannot be chi-square.
B. A t generator requires a uniform numerator.
C. Dividing the sum of squares by 6 is incorrect.
D. The numerator is not independent of the chi-square denominator.

2d)

The target is a sample of **independent observations** from a mixture of two **bivariate** normal distributions, with probabilities 0.30 and 0.70. How should the component label be generated for each observation?
A. Choose independent labels for the two coordinates.
B. Choose one label for the whole vector, then draw that vector from the selected component.
C. Draw a vector from each component and return their weighted average.
D. Choose one label once and use that component for every observation in the dataset.

2e)

Suppose $\Lambda \sim \text{Gamma}(3, 2)$ with rate 2, and $Y \mid \Lambda \sim \text{Poisson}(\Lambda)$. What is the **marginal variance** of Y?
A. 2.25
B. 1.50
C. 0.75
D. 3.00

2f)

Each row of an $n \times 2$ matrix Z contains two independent standard normal draws. Let

$$R = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}.$$

If `X <- Z %*% R`, what is the covariance matrix of a row of X?
A. $\begin{pmatrix} 5 & 1 \\ 1 & 1 \end{pmatrix}$
B. $\begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$
C. $\begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix}$
D. $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$

2g)

To generate a pair of independent standard normals, a programmer draws one $U \sim \text{Unif}(0, 1)$ and returns $X_1 = \Phi^{-1}(U)$ and $X_2 = \Phi^{-1}(U)$, where Φ is the standard normal CDF. Which conclusion is correct?
A. The pair is correct because both marginal distributions are standard normal.
B. Both marginals are correct, but the coordinates are equal and hence not independent.
C. The coordinates are independent because the quantile function is deterministic.
D. Neither coordinate has a standard normal distribution.

2h)

For independent replications with the same finite-variance model, an estimator's Monte Carlo standard error is approximately proportional to $1/\sqrt{B}$. To reduce it from 0.03 to 0.01, approximately how many replications are needed relative to the current number?
A. One third as many
B. Three times as many
C. Six times as many
D. Nine times as many

Question 3: True or False [6]

Write **True** or **False** for each statement. Each statement is worth **1 point**. No explanation is required.

3a)

Sampling all n observed records without replacement only permutes the records, so their sample mean is unchanged.

3b)

If X has a discrete CDF F, then $F(X)$ is necessarily uniform on $(0, 1)$.

3c)

If independent gamma variables have the same rate, their sum is gamma with the sum of their shapes and that common rate.

3d)

In R, `rnorm(100, mean = 0, sd = 2)` targets a normal distribution with variance 2.

3e)

In rejection sampling, replacing a valid bound M by a larger valid bound preserves the target distribution of accepted values, although acceptance becomes less frequent.

3f)

Matching a target distribution's mean and covariance is sufficient to establish that a multivariate generator has the correct joint distribution.

Question 4: Explain What the Results Mean [8]

Each part is worth **4 points**: **1 point for each numbered prompt**. Use one or two sentences per prompt. We grade the statistical and computational reasoning, not the exact wording of your response.

4a) A mixture is not a weighted average [4]

The target is

$$0.25\,N(-2,1) + 0.75\,N(2,1),$$

where the second parameter of each normal distribution is its **variance**. A student instead generates independent $A \sim N(-2,1)$ and $B \sim N(2,1)$ and returns

$$W = 0.25A + 0.75B.$$

Explain:

1. Describe a correct procedure for generating one observation from the target mixture.
2. Find the mean of the target mixture and $E(W)$. Would checking only the mean detect the error?
3. Find the variance of the target mixture and $\mathrm{Var}(W)$.
4. Suppose the two sets of simulated values are each recentered and rescaled to have mean 0 and variance 1. Does this establish that they have the same distribution? Give one additional distributional check.

4b) Simulation precision and generator error [4]

The intended model is an exponential distribution with **rate 1**. The exact probability of interest is

$$p_0 = P(X > \log 5) = 0.20.$$

A claimed generator is run for $B = 10{,}000$ independent observations. Exactly 2,400 observations exceed $\log 5$, giving $\hat{p} = 0.24$. All logarithms are natural.

Explain:

1. Compute the plug-in MCSE, $\sqrt{\hat{p}(1-\hat{p})/B}$, to four decimal places.
2. Express $|\hat{p} - p_0|$ in units of this MCSE. Is the discrepancy small enough to dismiss as ordinary Monte Carlo variation, or does it warrant investigating the generator?
3. If the same algorithm is run for 40,000 independent observations and $\hat{p}$ remains close to 0.24, what approximate MCSE would you expect?
4. Name one concrete implementation or parameterization error worth checking. Explain why increasing B would not correct that error.